Microsoft Elevate

CAPSTONE PROJECT

Cybersecurity Breach Analytics (2004 – 2025)

PRESENTED BY

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OUTLINE:

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PROBLEM STATEMENT:

- **The Scale of Loss:** Over 18.5 Billion records have been compromised since 2004, yet organizations lack a unified way to visualize these macro-trends.

- **Complexity:** Raw breach data is often cluttered with irrelevant metadata (URLs, stories), making it difficult for stakeholders to identify primary attack vectors.

- **Critical Need:** Without interactive analytics, security teams cannot accurately predict which sectors (Healthcare, Tech, Govt) are currently at the highest risk.

PROPOSED SOLUTION:

- The proposed system aims to address the challenge of visualizing and analyzing global data breach trends to enhance situational awareness for cybersecurity professionals. The solution will consist of the following components:
- **Data Collection:**
  - Gather historical data on cybersecurity breaches, including organization names, total records lost, year of occurrence, and industry sectors.
  - Utilize diverse data attributes such as data sensitivity scores and specific attack methods to categorize breach severity.
- **Data Preprocessing:**
  - Clean and preprocess the collected data to handle missing values, such as removing irrelevant "story" text and broken URL links.
  - Normalize "Records Lost" numerical values into millions and billions for consistent cross-visual comparison.
- **Data Analytics Logic:**
  - Implement **Top-N ranking logic** to automatically identify the top 5 most impactful breaches based on record volume.
  - Apply categorical grouping (Data Binning) to condense smaller industry sectors into an "Other" category for visual clarity.
- **Deployment:**
  - Develop a professional, interactive **Power BI Dashboard** featuring a dark-themed user interface for better data visibility.
  - Deploy the solution with integrated global slicers, allowing users to filter the entire threat landscape by specific years or sectors in real-time.
- **Evaluation:**
  - Assess the dashboard's effectiveness by measuring the time required to identify the primary attack vector for any given year.
  - Fine-tune visual interactions and DAX measures based on user feedback to ensure high data accuracy and readability.

SYSTEM APPROACH:

- **System Requirements:** Microsoft Power BI Desktop on Windows 10/11 with 8GB+ RAM for processing large-scale data aggregations.

- **Data Transformation:** Used **Power Query** for advanced cleaning to filter noise and normalize records into millions/billions.

- **Analytical Logic:** Implemented **DAX (Data Analysis Expressions)** to calculate dynamic KPIs like "Avg. Severity Score".

- **Visual Interface:** Built a high-contrast dark-mode dashboard with **Global Slicers** for real-time temporal analysis.

ALGORITHM & DEPLOYMENT:

- **Algorithm Selection:** Utilized **Top-N Ranking Logic** to isolate the top 5 most impactful breaches based on record volume.

- **Data Aggregation:** Applied **Statistical Measures** (Sum/Average) to calculate core KPIs like "Total Records Compromised" and "Avg. Severity Score".

- **Data Input:** Integrated historical logs containing organization names, years, attack methods, and record counts.

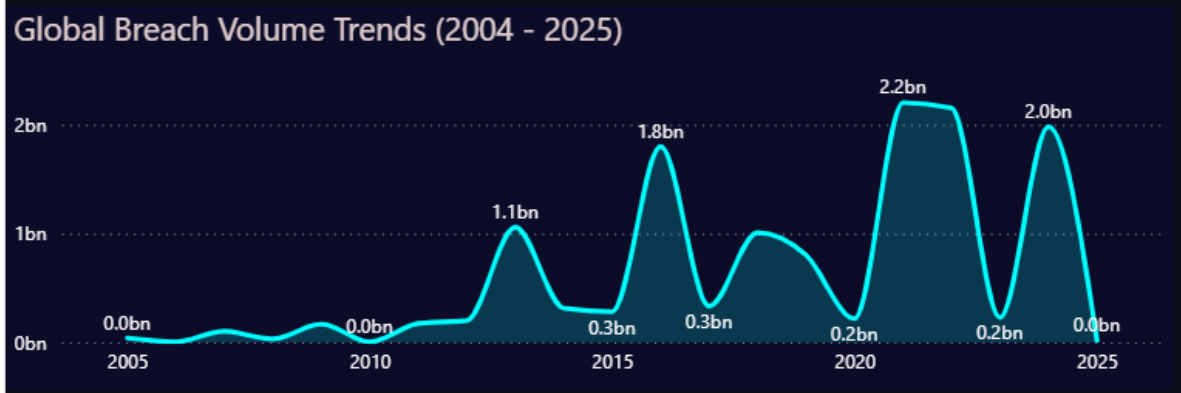
- **Deployment:** Developed as an interactive **Power BI dashboard** with global year slicers for real-time situational analysis.

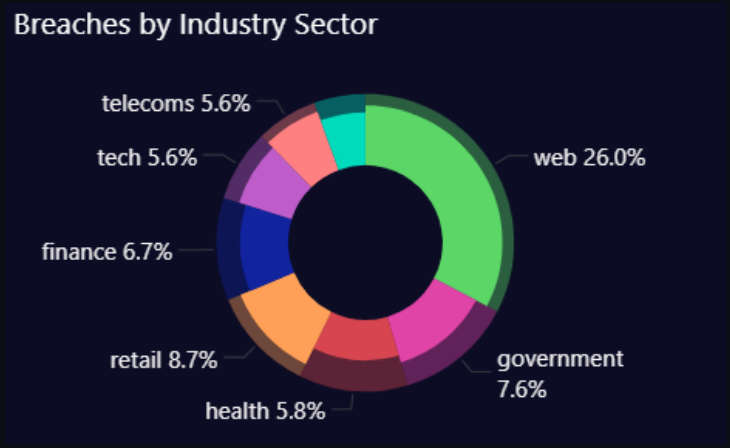
RESULT:

GLOBAL CYBERSECURITY BREACH ANALYTICS

Select Year: All

13.1bn	365	2.3	hacked
Total Records Compromised	Total Breach Incidents	Avg. Severity Score	Primary Attack Vector





organisation	Year	Records lost
Yahoo	2013	1,082M
Linkedin	2016	817M
Facebook	2018	612M
Indonesian SIM cards	2022	1,000M
National Public Data	2024	1,000M

CONCLUSION:

- **Effectiveness:** The dashboard successfully converted 20 years of "noisy" breach data into 4 critical KPIs and 3 interactive charts.
- **Key Finding:** Hacking remains the dominant threat vector, with the "Web" sector being the most frequently targeted industry.
- **Importance:** Accurate visualization of breach patterns is essential for proactive resource allocation in cybersecurity.

FUTURE SCOPE:

- **AI Integration:** Incorporating predictive machine learning models to forecast the probability of breaches based on emerging ransomware cycles.

- **Real-time Feeds:** Connecting the dashboard to live CVE (Common Vulnerabilities and Exposures) databases for immediate threat alerting.

- **Regional Expansion:** Expanding the dataset to include granular geographic data for localized threat assessment.

REFERENCES:

GitHub Link: https://github.com/AryanAnand-ux/Global-Cybersecurity-Breach-Analytics-PowerBI.git

Thank You